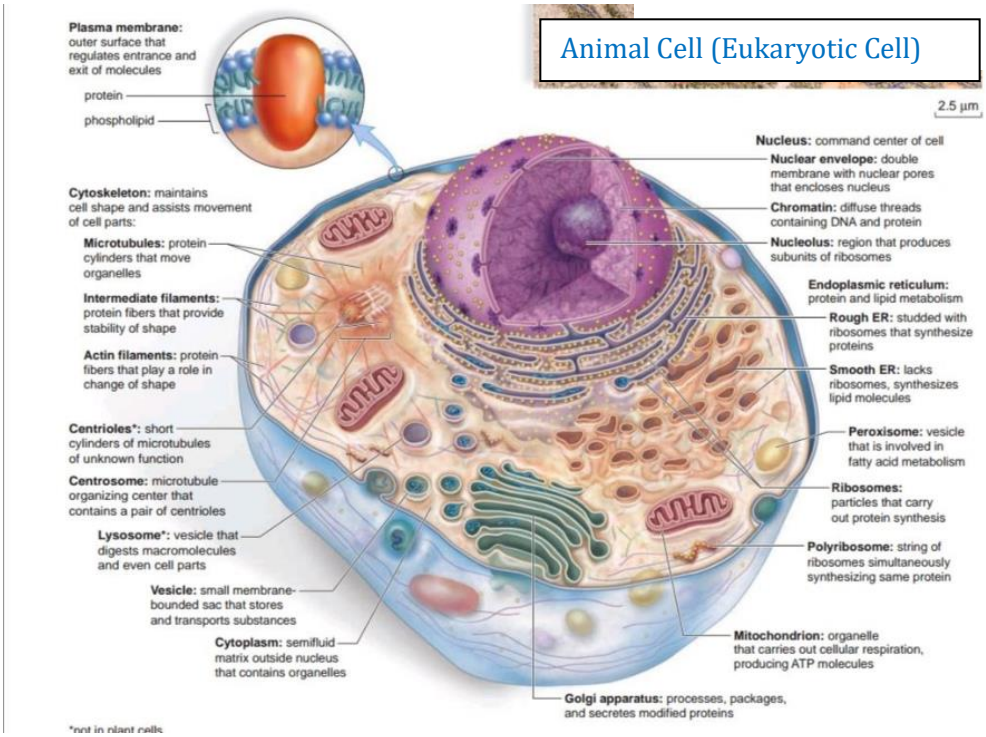
Medicine, Dentistry and Medical laboratory

Cell Biology

Lecture 4

Cell structure & function - 2

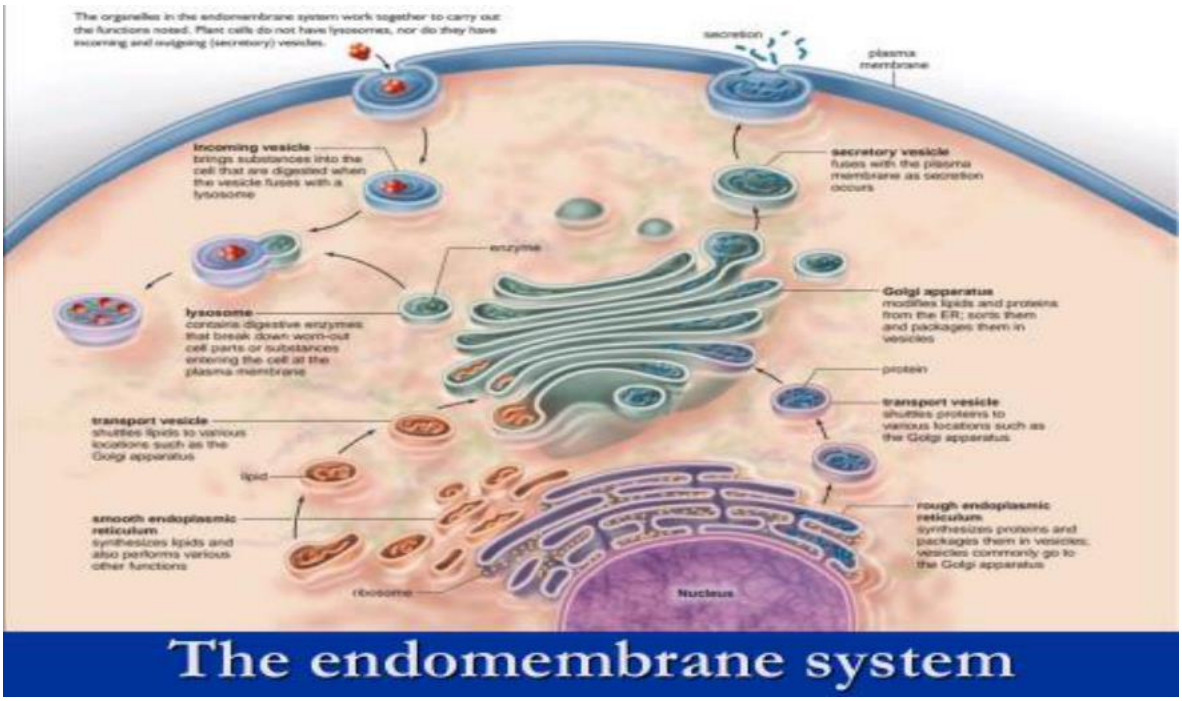
A general living cell (e.g. animal cell) is composed of:



4- **Lysosomes** are simple spherical sacs (bud off the Golgi apparatus in cells) contains digestive enzymes that digest worn-out cell parts and substances entering the cell at the plasma membrane.

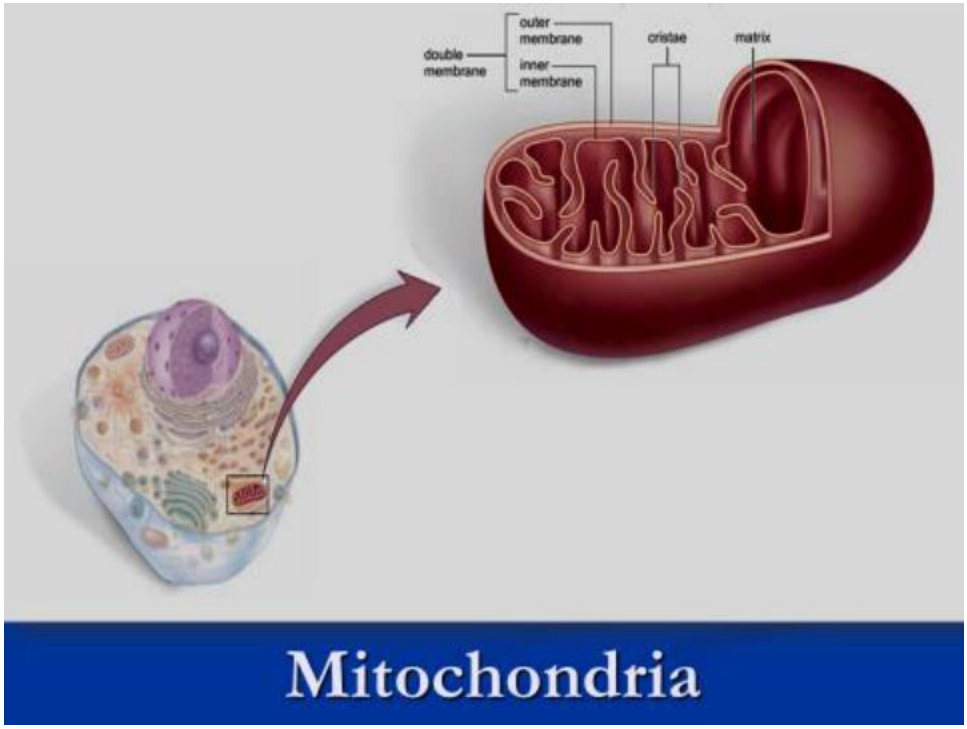
* **The endomembrane system consists of:**
The nuclear envelope, the membrane of the endoplasmic reticulum, Golgi apparatus and several types of vesicles

* The endomembrane system is a series of membranous organelles that work together and communicate by means of transport vesicles.



5-Mitochondria:

Are composed of two membrane (inner and outer), small piece of circular DNA, ribosomes and respiratory enzymes.

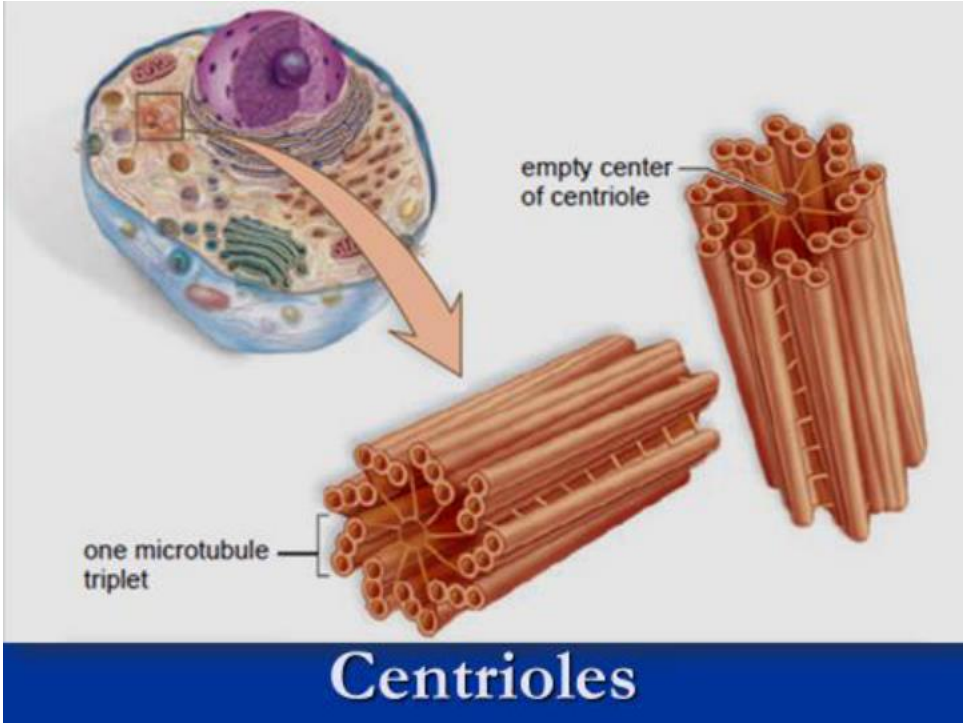


Function:

* Mitochondria are involved in cellular respiration.
* Mitochondria are often called the powerhouses of the cell because they produce most of the ATP utilized by the cell.

6- Centrioles:

A pair of hollow cylinders arranged at right angles (found in animal cells only). Each pair is composed of 9 triplets of microtubules.
Function: formation of spindle fibers during cell division.



7- Cilia: are hair-like projections composed of 9+2 arrangement of microtubules.

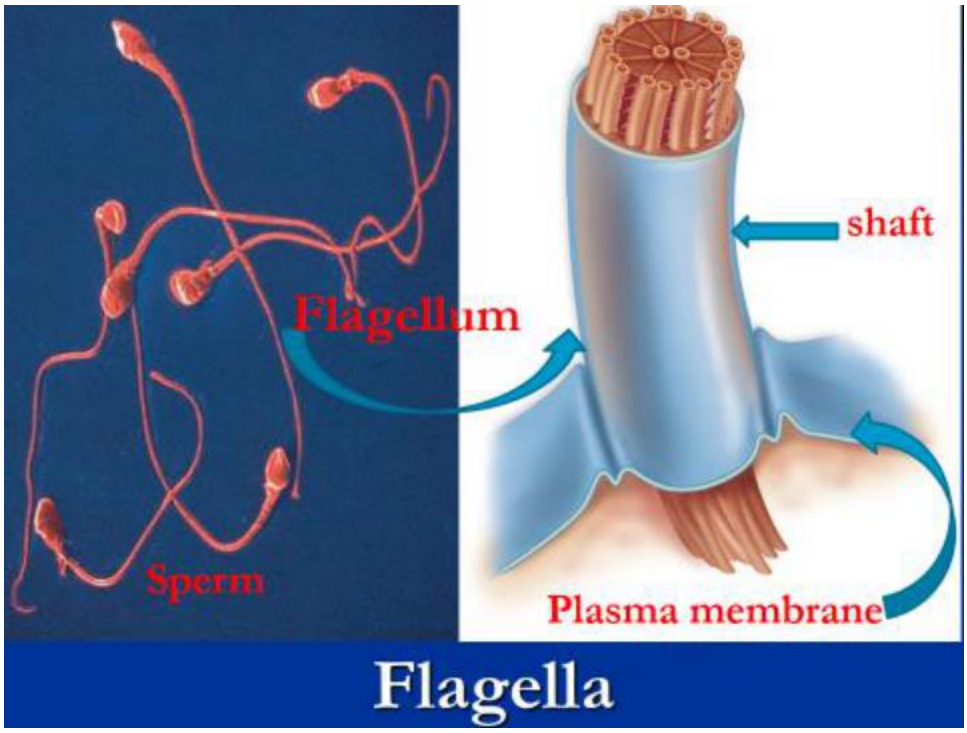
* Cilia move the cell through
liquid (e.g. unicellular paramecia are organisms that move by means of cilia).

* The cells that line our upper respiratory tract have cilia that sweep debris trapped within mucus back up into the throat, where it can be swallowed. This action helps keep the lungs clean (protection).

8- Flagella:

The shaft of the flagellum has a ring of nine microtubule doublets anchored to a central pair of microtubules.

Function: locomotion (move the cell through liquid).



9- Vacuoles:

***** Like vesicles, vacuoles are membranous sacs, but vacuoles are larger than vesicles. Few animal cells contain vacuoles.

***** Fat animal cells contain a very large lipid-engorged vacuole that takes up nearly two-thirds of the volume of the cell!

Function:

* Contractile vacuole: Osmoregulation.

* Food vacuole: Nutrition.

10- Peroxisomes:

Peroxisomes, similar to lysosomes, are membrane-bounded vesicles that enclose enzymes.

* The enzymes in peroxisomes are synthesized by free ribosomes and transported into a peroxisome from the cytoplasm.

* All peroxisomes contain enzymes whose actions result in hydrogen peroxide (H_2O_2):

$RH_2 + O_2 \longrightarrow R + H_2O_2$.

* Hydrogen peroxide, a toxic molecule, is immediately broken down to water and oxygen by another peroxisomal enzyme called catalase.

Function:

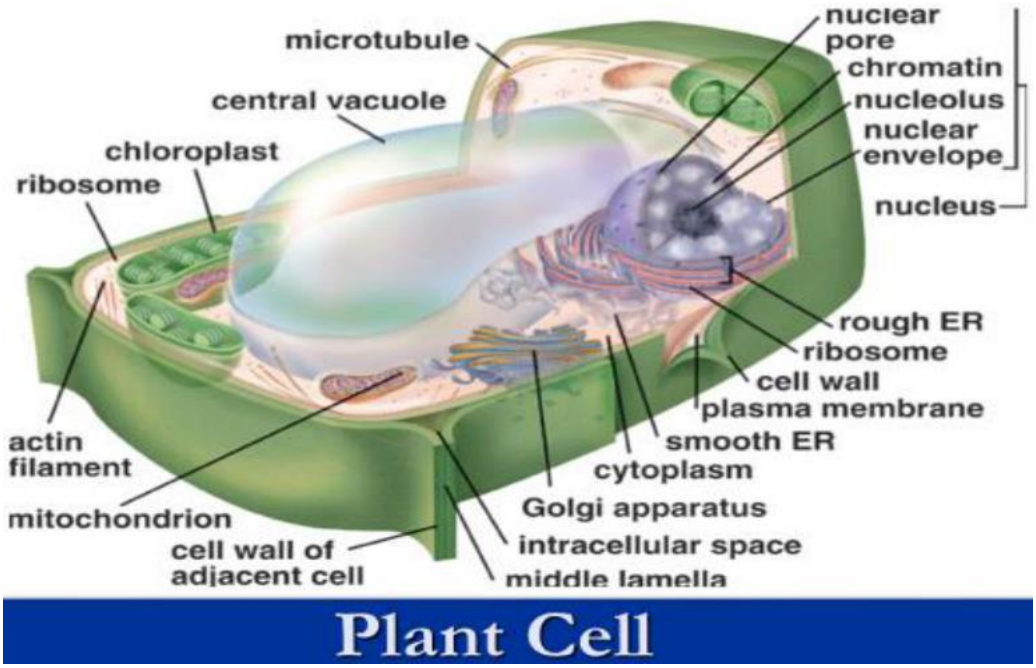
Prevalent in cells that are synthesizing and breaking down lipids. In the liver, some peroxisomes produce bile salts from cholesterol, and others break down fats.

*** A general living cell (e.g. plant cell) is composed of:**

Plant cell structure:

In addition to all structures described in animal cell (except few structures), plant cell is composed of:

1- Cell wall. **2- Central vacuole.** **3- Chloroplast.**



1- Cell wall:

***** Is different in its composition from that of the fungi and prokaryotes.

***** It is a dead part of the cell which is secreted by the living cell.

***** The cell wall consists of cellulose on a matrix of other polysaccharides.

* It contains about 60-70% water.

Function: provides mechanical support and protection.

2- Central vacuole:

A vacuole is a fluid filled sac bound by a single membrane. It is surrounded by a tonoplast (membrane) and it contains the cell sap.

*** The cell sap:**

It is concentrated solution of minerals, sugars, organic acids, oxygen, CO_2, pigments and some waste.

Function:

***** Storage of various substance e.g. food (sucrose) or waste (crystals and latex).

***** Osmotic uptake of water during osmosis which is important in cell expansion during cell growth.

3- Chloroplast:

It is a large plastid which is surrounded by an envelope of two membranes, it contains a system of membranes (thylakoids) which runs through a ground substance called stroma.

Function:

* The organelle in which photosynthesis occurs producing sugars from CO_2 and water using light energy trapped by chlorophyll (energy converter).